

EDUSENSE: A MULTIMODAL LEARNING ASSISTANT FOR SPECIALLY-ABLED STUDENTS

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Abstract— For many students, the traditional classroom isn't just a place to learn it's where they constantly face obstacles. Imagine a child trying to keep up with a fast-moving lesson through sign language, or a student with autism feeling overwhelmed by noise and distractions. The usual teaching methods often don't meet their needs. That's why we created EduSense to bridge these exact gaps. It translates hand gestures into real-time text, helping hearing-impaired students communicate easily with their classmates. We've also introduced gesture-based learning modules that offer students on the autism spectrum a calm, structured, and stress-free way to participate. Even something as simple as turning a physical textbook into digital content can make a huge difference for a student with dyslexia. Our aim isn't to build flashy "assistive technology," but to create a tool that truly helps every child feel included and supported.

Keywords—Multimodal learning, Special education, Sign language, Deep learning, Assistive technology, Emotion recognition, Behavior profiling, Dyslexia support, Inclusive education.

I INTRODUCTION

Education is meant to level the playing field, but for so many kids with disabilities, traditional classrooms are packed with hidden hurdles. You can picture a hearing-impaired student struggling to follow along with spoken lessons, or a child with autism totally overwhelmed by noise and distractions. The typical "one-size-fits-all" teaching method often falls short for those who need that extra support. Sure, there are assistive tools out there, but most zero in on just one issue, leaving teachers to scramble for a way to make the space truly inclusive. That's exactly why we created EduSense. It flips the script with multimodal AI that shapes itself around the student, instead of making them bend to the technology.

EduSense brings together real-time sign language translation, straightforward gesture-based interactions for neurodivergent learners, and OCR tech to turn printed materials into simple digital content—all in one easy platform. Throw in interactive games, and learning isn't just doable; it's fun and keeps every kid motivated. With EduSense, we're showing how computer vision and machine learning can feel genuinely human-centered. By letting students learn their way—through signs, gestures, or digital text—we're building classrooms where no child gets left behind, no matter how they communicate or take in the world.



Figure 1. Representation of Specially-abled Persons

This picture explains how EduSense helps students feel comfortable in a digital classroom. It does not follow the same learning method for everyone. Instead, it gives different kinds of support depending on the student. Some learners use visual help, some use audio options, and others use simple hand gestures to learn better. The illustration also shows main features like live sign language conversion, special learning support for children with autism, and tools that help students who have difficulty in reading. All these are included in one simple system that is easy to use.

II LITERATURE SURVEY

In 2023, H. Terada [7] and his team studied how artificial intelligence can be used to support students with learning disabilities. They reviewed many AI-based systems that help in improving reading and basic math skills. The study showed that personalized AI tools give better learning results. However, most of the available tools focus on solving only one problem at a time and do not provide complete support for all learning needs.

S. M. Razak et al. [8] in 2023 analyzed different assistive technologies used in higher education for students with disabilities. They reviewed 133 research papers from IEEE Xplore and Scopus databases. From this study, they created a framework to classify inclusive educational tools. The authors mentioned that most current systems are separate and not combined into a single platform that can support multiple disabilities together.

In 2022, V. Rajesh and his team[4] developed a system using deep learning and IoT for individuals with Autism Spectrum Disorder. The system used wearable sensors connected to a Raspberry Pi to track heart rate and skin response to detect emotional stress. The results were effective in monitoring behavior, but the system mainly focused on stress detection and did not help with academic learning activities.

In 2024, P. Kumar and S. Gupta developed a real-time Indian Sign Language recognition system using convolutional neural networks. The system captured hand gestures from live video and converted them accurately. This helped improve communication for deaf and mute individuals, but it was mainly designed for communication purposes and not for classroom learning.

R. Sharma and N. Verma in 2024 proposed LexiEase AI to support students with dyslexia. The system used language processing and speech technologies to detect reading difficulties and simplify learning content. It also helped students with writing tasks. However, the system worked as a standalone tool and was not integrated with other learning platforms.

III METHODOLOGY

The proposed EduSense system is developed to support students with different learning needs by combining several assistive technologies into one platform. The methodology mainly focuses on collecting various forms of user input, processing them using intelligent models, and providing suitable outputs based on individual requirements. Although each module performs a specific task, they are interconnected through a common framework to maintain consistency and track student progress.

A. User Authentication and Profiling

User authentication ensures that only authorized students can access the platform. After login, a user profile is created for every student. This profile stores basic information along with learning preferences, activity records, and performance results. Over time, the system analyzes this data to modify lesson difficulty and content type according to the learner's improvement level. This helps students learn at a comfortable pace.

B. Sign2Talk: Gesture Recognition Module

The Sign2Talk module is designed to help students who rely on sign language for communication. A webcam captures hand movements, and important features such as finger positions and hand orientation are extracted. These features are then processed using a deep learning-based recognition model to identify the corresponding sign. To avoid errors, continuous frame analysis is used so that sudden or unclear gestures do not affect the output.

C. TouchRead: Reading Support System

TouchRead focuses on assisting students who experience difficulty in reading printed or digital text. When a student points at text, the system uses Optical Character Recognition (OCR) to convert the image into editable digital text. This text can be displayed in a simpler form or converted to speech. This allows learners to read at their own speed and improves understanding.

D. Gamified Learning Approach

To make learning more engaging, EduSense includes game-based activities. Educational tasks are presented in the form of small challenges where students earn points or visual progress indicators based on their performance. The system records accuracy and completion time to evaluate improvement. This approach helps in maintaining student interest while promoting continuous learning.

E. Output Accessibility Module

This module provides multiple output formats to suit different learner needs. The Text-to-Sign feature converts written content into sign language animations, making it easier for hearing-impaired students to understand lessons. The Text-to-Speech feature reads out the text for students who struggle with reading. By offering both options, the system ensures better accessibility.

F. Interactive Reading using OCR and Finger Tracking

This part of the system allows students to interact directly with real-world text. OCR is used to scan text from books or screens, while finger tracking identifies the word being pointed at. The system then highlights or reads the selected word. This method makes reading more interactive and easier for students to follow.

EduSense is designed by bringing different support tools together into one learning system. It includes features such as gesture recognition, reading assistance, and simple game activities to help students with various learning difficulties. Instead of using the same content for everyone, the system creates personal profiles and adjusts lessons based on how each student performs. Tools like OCR and finger tracking help students interact easily with books and other materials, while deep learning improves the accuracy of recognizing hand gestures. In general, EduSense provides a comfortable and inclusive learning space that helps students learn better and feel more confident.

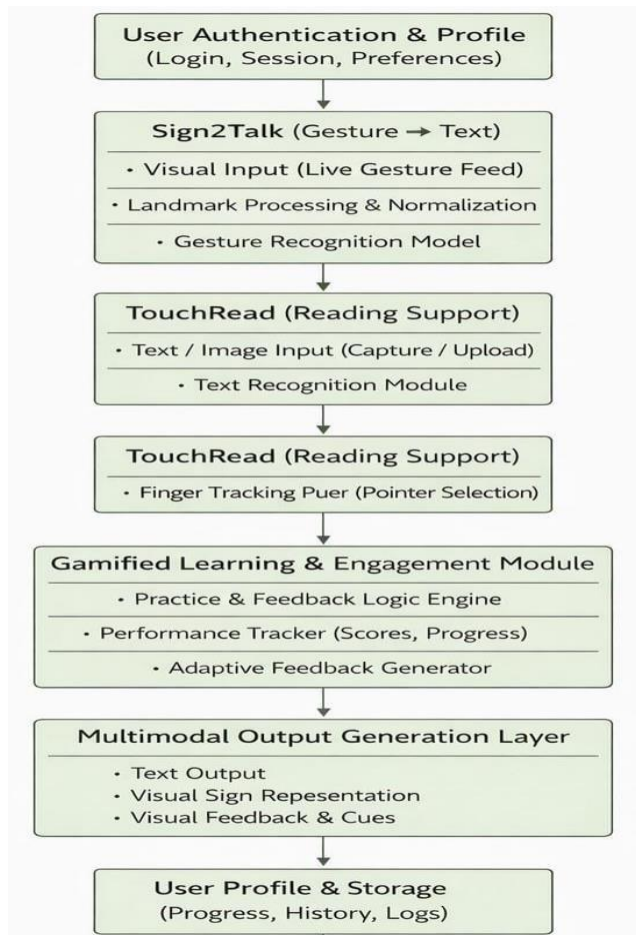


Figure2. Proposed System Architecture

IV RESULTS AND DISCUSSION

The EduSense system was tested using the developed web application to understand how each feature works in real-time situations. The evaluation was done by observing system responses, ease of use, and how helpful each module was for specially abled students. The performance of every module was reviewed separately.

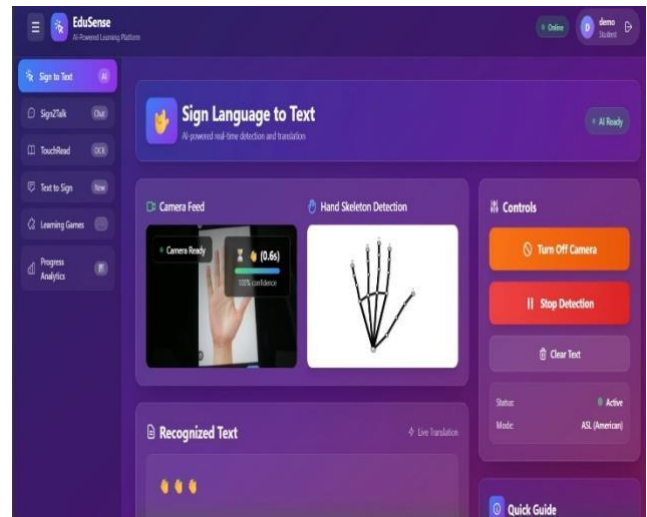


Figure3. Sign Language to Text

Figure 3 shows the interface where hand gestures are captured using a camera. The live video and detected hand structure are displayed, and the recognized signs appear as text. From the output, it can be seen that hand movements are detected properly and finger positions are tracked continuously. The system converts gestures into text without noticeable delay, which makes communication smooth. The hand structure view also helps users position their hands correctly. This module helps hearing and speech impaired students communicate easily using gestures.

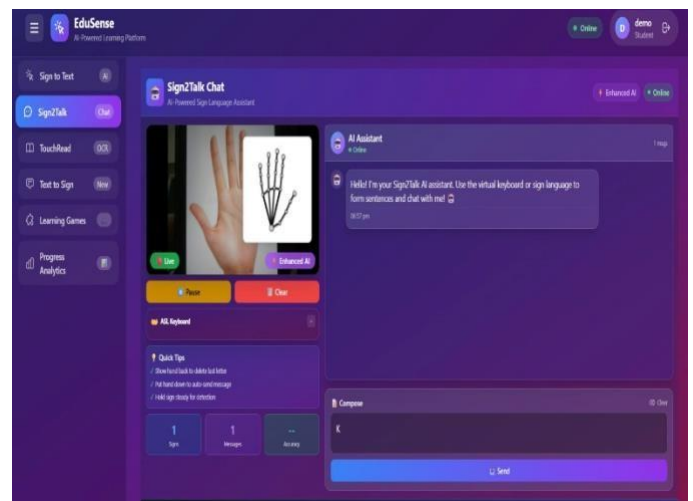


Figure4. Sign to Talk

Figure 4 displays the chat interface where gesture inputs and text messages are used together to form conversations. The chat system allows users to move from single gestures to full message communication. The layout is simple and easy to understand. This shows that EduSense can support normal conversation along with learning activities, making it useful in classrooms.



Figure5.Touch Read

Figure 5 presents the TouchRead interface with camera view and recognized text. The system is able to detect printed text and link it with finger position accurately. This helps user focus on specific words while reading. The method reduces confusion and makes reading easier, especially for students with learning difficulties.

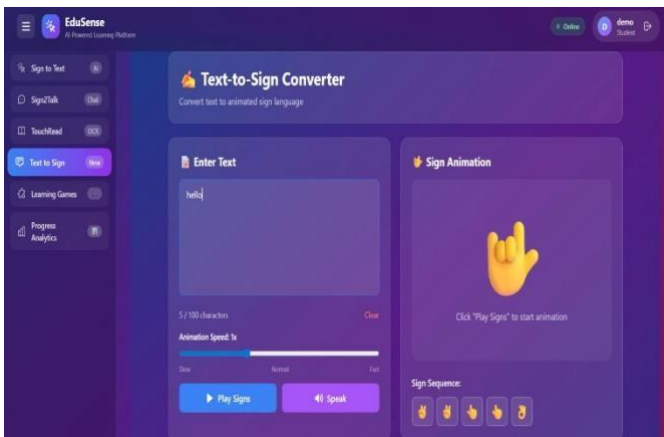


Figure6. Text to Sign Language

Figure 6 shows how entered text is converted into sign language visuals. This module helps convert written content into sign format for better understanding by hearing-impaired users. The step-by-step display of signs improves clarity. It supports two-way communication between different learners.

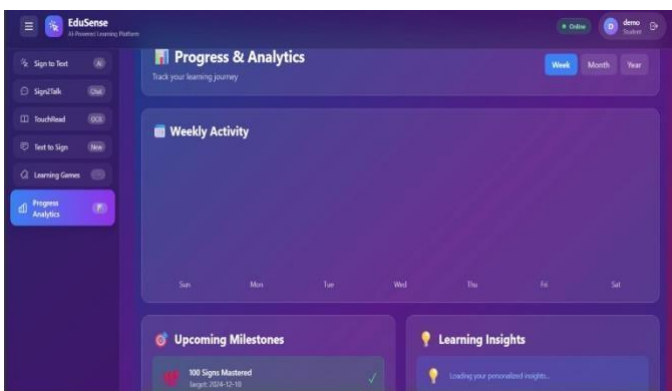


Figure7. Progress and Analytics Module

Figure 7 presents the dashboard showing user activity and learning progress. This module helps track student performance over time. It provides useful information to understand improvement levels and learning patterns. The data supports personalized content adjustments. The results across all modules demonstrate that EduSense functions as a cohesive multimodal learning assistant. The system maintains consistent performance during real-time interaction and allows seamless switching between modules. The integration of gesture recognition, reading assistance, bidirectional communication, gamified learning, and progress tracking validates the effectiveness of the proposed approach.

V CONCLUSION

In this project, we built EduSense, a learning assistant designed to help specially abled students by combining several helpful features in one platform. Unlike most tools that focus on just one problem, EduSense brings together sign language recognition, reading support, text-to-sign conversion, interactive learning activities, and progress tracking.

We tested the system across its main features, and it worked well. The sign language recognition allowed students with hearing or speech difficulties to communicate using gestures. The TouchRead feature guided fingers along the text, which made reading easier for those with difficulties. The text-to-sign conversion helped users communicate both ways. Adding game-like activities kept students interested and motivated. The progress tracking feature also gave useful information about learning and improvement.

Overall, EduSense showed that it can be a practical tool for students with different learning needs. By combining several accessibility features with personalized support, it helps make learning more inclusive. This project shows how putting multiple assistive tools together can make education more accessible and useful for specially abled students.

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